

Universe24: one local integer rule on the cube's lattice, the click as its one non-local act, and the universe it builds without a free number

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Abstract

Can one local rule in whole numbers carry the quantum and the relativistic forms? The law is one line, Rule3: at every Node, each interval, a second-order step of a family's level from its two levels and its six neighbours' arrivals, with integer coefficients from the family's pair and the Node's paces, and one Euclidean division whose remainder stays at the Node. A quantum moves whole when a remainder crosses the quantum's action: the count's line, Rule3 read on its own conserved current, whose division is the click, the one non-local act. Exact on the lattice: reversibility bit for bit, a conserved form, the count's conservation, one quantum per click, Born's rule, no signalling, and for a pair record the correlation $\cos 2(a - b)$ with the Clauser-Horne-Shimony-Holt sum $2\sqrt{2}$. The families are bands of the rule: three exact rotations of periods 6, 4 and 3, three tensor ranks from the three reads of the line, one free number per rank. A universe of whole bodies is declared: a body is the fixed point of its binding row on its Nodes, and the count moves by the count's line; one hollow alone has a saddle and no stable extended body, and the hill that holds a body is a hypothesis. The forms of Einstein, Lorentz, de Broglie, Born, Malus and Bell are derived from the line as predictions, and every experiment is run algebraically on a declared world; nothing is run on the engine and no agreement with nature is claimed.

Keywords: integer lattice model; one local rule; the click as the measurement; Bell correlations; discrete relativity; pre-registered simulation

1 Introduction

The question. The law of this paper is one line of integer arithmetic applied at every Node of a cubic lattice at every interval. Nothing else acts: no field equation beside it, no law of motion for a body, no rate, no draw, no real number. The paper states that line, proves what is exact on the lattice, derives what follows from the line alone, names what is assumed beside it, and writes down, before any run, what eighteen experiments read in clicks. The forms of Newton, Einstein, Lorentz, de Broglie, Born, Malus and Bell appear only on the comparison side, beside the law's own expectation; none is an input.

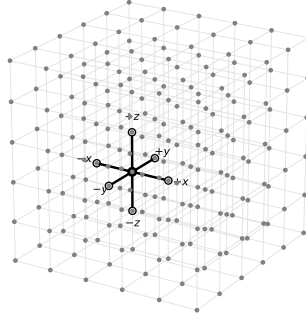
The law in one line. At every Node, each interval, for every family:

Rule3:	$w a_{\text{next}} + r' = \sum_{a \in \{x, y, z\}} R_a (\text{arr}_{+a} + \text{arr}_{-a}) + S a_{\text{now}} - w a_{\text{before}} + r, \quad 0 \leq r' < w,$ $w = 6 \text{den} \Gamma^2, \quad R_a = 2 \text{num} p_a^2,$ $S = 12 \text{den} \Gamma^2 - 12 (\text{den} - \text{num}) p_0^2 - 4 \text{num} (p_x^2 + p_y^2 + p_z^2);$	(1)
the count's line:	$T c_{\text{next}} + r' = T c_{\text{now}} + \sum_j F_{ij} + r, \quad 0 \leq r' < T,$ $F_{ij} = \text{weight} \times (\text{now}_i \text{ before}_j - \text{before}_i \text{ now}_j);$	
the one write:	$\text{a level at a Node gains } (w q(\text{Node}) + r) \text{ div } E, \text{ the remainder } r \text{ kept there.}$	

In the first line a_{now} , a_{before} and a_{next} are the family's level at the Node now, one interval before and one interval on; $\text{arr}_{\pm a}$ is the arrival through the Port $\pm a$, the neighbour's level or its rotation by the Link's angle; r and r' are the remainder kept at the Node before and after the step; Γ is the Node clock, the pace of an empty Node; $[\text{num}, \text{den}]$ is the family's pair, with $\cos \omega_0 = \text{num}/\text{den}$ its rotation per interval at wave number zero; p_0 is the Node's pace and p_a its pace along the axis a , both the Node clock lowered by the levels the family reads there; and div is the floor division. The second line is the same rule for the family of clicks: c is the count, the number of quanta at the Node; T is the quantum's action, the one unit of every family; F_{ij} is the current into the Node i from its neighbour j over the interval, the bilinear form of the record's two levels at the two ends; and the division is the click, one quantum moving whole exactly when the remainder crosses T . The third line is every write of a body or a record into a level: $q(\text{Node})$ is the writer's own quantity there, w the one weight with which the writer's family reads the written family, and E the written row's divisor. Everything else is a declared number. Section 2 states the first line with its proofs, Section 3 the second and Section 4 the third; the paper cites the law's document [1] by the name of the section each formula stands in.

The seven things, in the order they are built. Each is named once, and the words are the law's [1, the objects].

- (1) *The GameBoard.* A box of Nodes in $\mathbb{Z}^3$, each joined to six neighbours by Links through its six Ports $\pm x, \pm y, \pm z$; each axis periodic or open. The whole local information of a Node is its NodeState: the levels and remainders of the families there, and nothing else.
- (2) *The family and the record.* A family is one kind of physical value with its own declaration: its pair, its parts, its phase, what it reads and what it holds. Its record is a set of integer levels over the GameBoard, one per Node per level, with a remainder per Node; a family of two levels per Node holds a pair (re, im).
- (3) *Rule3.* The first line of Eq. (1), the one LocalRule: every family at every Node steps by it and by nothing else, a body's own record included.
- (4) *The click.* The second line: the family of clicks is a family like every other, its level the count, its wall T , its read the current; its division is the click, the law's one non-local act, at which a record ends whole where its last quantum leaves it.
- (5) *The one write.* The third line, in five instances: the hold, the source, the giving, the start and the recoil.
- (6) *The body.* A body is its count and its record: a set of Nodes on which a family's count stands, its record the standing wave the count binds, its momentum and its spin readings of that record's current. A body has no law of its own.
- (7) *The universe and the world.* Every physical number is a declaration outside the rule: the universe's, the families' rows and the universe's integers, and the world's, the bodies by their Nodes and counts, the detectors and the readings. The rule holds no number, no family's name and no exception.



one Node and its six Ports, the Links to its six neighbours

Fig. 1 The GameBoard in space: a cubic array of Nodes, here $6 \times 6 \times 6$, each joined to its six neighbours by Links through the Ports $\pm x, \pm y, \pm z$ (the faint lines); one Node and its six neighbours marked, the Nodes one interval away from it, whose convex hull is the octahedron of Fig. 2. A Node keeps each family's two levels and its remainder there, and nothing else. Drawn from the definitions; no run.

The tree: what needs what. The paper's claims are sorted by what they need beyond the line, and each carries one of five marks: THEOREM, proved here for every GameBoard; DERIVED, from theorems by argument; COMPUTED, a number of the law's own computed before any run; ASSUMPTION, named where it is used; HYPOTHESIS, stated under its own name [1, the families from the rule, the tree]. The root is the line, Section 2. From it alone: the reversibility and the conserved form (Theorems 1 and 3), the guard's edge (Proposition 3), the count's conservation, one quantum per click, Born's rule and the pair's correlation $\cos 2(a - b)$ (Theorems 4, 5, 6 and Proposition 4, Section 3), the three exact rotations and the three reads of the line with the ranks 1, 3 and 6 (Propositions 5 and 6, Section 6). With the assumption that the fields every family reads are the fields of unbounded reach: one free number per rank, DERIVED (Section 6.3). With the owner's word that no body is one Node: the universe of whole bodies, Section 7, its declaration, its generator as the fixed point of the binding row, the shape of a symmetric well's bound record (Proposition 8) and the scaling of a hollow (Proposition 9, DERIVED), with a run's finding that one hollow's fixed point is a saddle; under a hypothesis in its own name, the hill that holds a body. Under the hypothesis of the divisor 1, the cluster universe of one-Node bodies with its edge 0.2255Γ (COMPUTED) and horizon $\Gamma \div 2$ (DERIVED), which holds no extended body. With the assumption that every body is bound: the click that joins and parts, the observer as a body. Under hypotheses in their own names: the third, the conversion, the electron as the lightest band, the growing Node clock. The forms of nature, Section 8, are DERIVED from the line as predictions, and every experiment is run algebraically on a declared world (Table 1, COMPUTED); none is run on the engine. Not closed toward nature by name: $\alpha, m_e/m_P, m_p/m_e$, the quarks' charges, the generations and the band $[0, 1]$ (Section 9).

The reading rule and the notation. Only a detector's click is a measurement; an experiment is compared with nature through its detectors' clicks alone [1, readings and measurements]. A number of the GameBoard's state (a level, a support, a total, a body's centre) is a diagnostic, labelled GAMEBOARD, never compared with nature and never pinned; a number computed from the algebra before any run is labelled COMPUTED; a declared integer of a world or a universe is a DECLARATION; a number of the bookkeeping outside the law is HOST. Every physics name in this paper (Born's, Malus's, Lorentz's, Tsirelson's) names the known form the law's result is set

beside. Every symbol is named in English at its first use: a scalar plain, a vector in bold lowercase ($\mathbf{n}$), a matrix or an operator in bold uppercase ($\mathbf{M}$). `div` and `mod` are the floor division and its remainder, `isqrt` the integer square root.

What this paper contributes, and what it does not claim. New here: one explicitly stated integer rule whose reversibility, conserved form, count conservation, one-click theorem, Born’s rule, no-signalling and pair correlation $\cos 2(a - b)$ are theorems on the lattice; the families as bands of that rule, with one free number per tensor rank; the universe of whole bodies with its generator, the scaling that makes one hollow’s fixed point a saddle, the hypothesis of the hill that holds a body, and the cluster universe of the divisor 1 as the hypothesis it replaces, each with its mark; and the forms of nature derived from the line as predictions, each beside the form it is compared with. Not claimed: any agreement with nature, since nothing is run here; a derivation of the Lorentz group, which is not a symmetry of the lattice (the lattice’s symmetries are the 48 signed permutations of the axes); the fine-structure constant, the electron’s mass ratios or the quarks’ charges, which are open by name; a stable extended body, which is the open hypothesis of the hill; or that nature is so. The paper is algebra: it holds no run, no code and no experiment, and the simulator that computes the rule is named only as the archive of the law’s document.

2 The GameBoard and Rule3

This section states the object and proves what is exact on it: the lattice, the families, the one line with its coefficients, its reversibility, what its acts can and cannot form, the paces and their guard, the transport and the conserved form. Every formula is the law’s [1, the GameBoard and the families; Rule3].

2.1 The objects

A *Node* is a physical location. The *GameBoard* is all the Nodes together, a box of shape $= (N_x, N_y, N_z)$ Nodes; each axis is periodic (the box a torus on that axis) or open (a face beyond which no Node lies), as the world declares. Two Nodes that differ by one step along one axis are joined by a *Link*; each Node has six Links, one through each of its six *Ports*, named by their outward direction $+x, -x, +y, -y, +z, -z$ (Fig. 1). An axis of one layer folds: its two Ports return the Node itself. The local information of a Node is its *NodeState*: the levels and remainders of the families at that Node, and nothing else. A change of a *NodeState* is an *Event*; the rule that makes it is a *LocalRule*, and there is one, Rule3.

A *family* is one kind of physical value with its own declaration. A family’s *record* is a set of levels over the GameBoard, one integer per Node per level, with a remainder per Node; a family with two levels per Node holds a pair (re, im) at each Node. A *body* is a set of Nodes on which a family’s *count* stands, with the values the body’s writer declares (its content, its momentum, its spin, its moment); a body is one connected region.

The universe’s integers. Every physical number is a declaration outside the rule: the universe’s (the families and the universe’s integers) and the world’s (the bodies, the detectors, the readings); the rule holds no number of its own [1, a family’s declaration]. The universe’s integers are the Node clock Γ , the pace of an empty Node and every family’s clock unit; the quantum’s action T , the action of one quantum of any family, a unit as Γ is; the momentum unit Q_{unit} ; the twist table (Section 2.7); and the least number of remainder values a giving family’s pair must leave at its giving Node, so that the residue of the ladder (Section 3.3) is spread over them. A family’s row

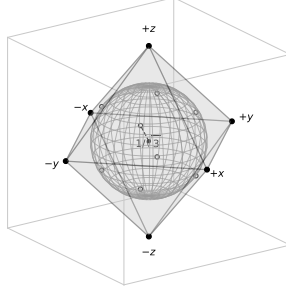


Fig. 2 The six Ports of one Node, $\pm x, \pm y, \pm z$, and the octahedron they span: the Nodes one interval away. The maps of the six Ports onto themselves that keep opposite Ports opposite are the 48 signed permutations of the axes, the symmetry group of the cube; the symmetric parts of a tensor under the 48 have the ranks 1, 3 and 6, the three ranks of Section 6. Drawn from the definitions; no run.

declares its pair $[\text{num}, \text{den}]$, written as one integer m over the GameBoard's one denominator Γ (Section 6.2), its parts (1, 3 or 6 components, the tensor rank of what writes it), its phase (2 where it carries quanta, 1 where it never clicks), the families it reads, and where it holds a body's count, the row's divisor E_s . The weights of every read are 1, the sign is the body's and the action of one quantum is T for every family, so the declarations hold Γ , T , the matter pair and the two divisors, and nothing else of physics.

2.2 The line

At every Node, each interval, for every family, with the wall w , the read coefficient R_a on each axis a , the self coefficient S , the level a_{now} at the interval's start, a_{before} one interval earlier and the remainder r kept at the Node,

$$w a_{\text{next}} + r' = \sum_a R_a (\text{arr}_{+a} + \text{arr}_{-a}) + S a_{\text{now}} - w a_{\text{before}} + r, \quad 0 \leq r' < w, \quad (2)$$

$\text{arr}_{\pm a}$ the arrival through the Port $\pm a$: the neighbour's level, or its rotation by the Link's angle (Section 2.7); 0 beyond an open face; the Node's own level on a folded axis. The remainder is the Node's own and never travels. The coefficients at a Node, from its paces (Section 2.5) and its family's pair, are

$$w = 6 \text{den} \Gamma^2, \quad R_a = 2 \text{num} p_a^2, \quad S = 12 \text{den} \Gamma^2 - 12 (\text{den} - \text{num}) p_0^2 - 4 \text{num} (p_x^2 + p_y^2 + p_z^2). \quad (3)$$

One floor division per Node per interval; the remainder stays at the Node. A held family's field (a real field of phase 1) steps at its row's pair with the pace 1 and the wall 3den , at first order $[1, \text{the line}]$.

Proposition 1 (the vacuum's wave rule and the band). *At the vacuum's pace, $p_0 = p_a = \Gamma$, the self coefficient vanishes and Eq. (2) is, up to the remainder, $a_{\text{next}} + a_{\text{before}} = \frac{\text{num}}{3 \text{den}} S_6(a_{\text{now}})$ with S_6 the sum over the six neighbours. A plane wave of wave number k along an axis rotates by ω per interval with*

$$\cos \omega = \frac{\text{num}}{3 \text{den}} (2 + \cos k), \quad \cos \omega_0 = \frac{\text{num}}{\text{den}} \text{ at } k = 0,$$

and off an axis $\cos k_x + \cos k_y + \cos k_z$ stands in place of $2 + \cos k$. The family's band of rotations is $[\text{num}/(3 \text{den}), \text{num}/\text{den}]$ in the cosine; light, the pair $[1, 1]$, has $\cos \omega = (2 + \cos k)/3$, its group

velocity at long wavelength $1/\sqrt{3}$ Links per interval, and its band reaches $\cos \omega = 1$; a pair with $\text{den} > \text{num}$ has the gap $\cos \omega_0 = \text{num}/\text{den}$ below 1, a rest rotation, a mass.

Proof. With $p_0 = p_a = \Gamma$, $S = 12 \text{ den } \Gamma^2 - 12 \text{ den } \Gamma^2 + 12 \text{ num } \Gamma^2 - 12 \text{ num } \Gamma^2 = 0$ and $R_a/w = \text{num}/(3 \text{ den})$. For $a_{\text{now}} = A \cos(kx - \omega t)$ along the axis x , $S_6(a_{\text{now}}) = (4 + 2 \cos k) a_{\text{now}}$ and $a_{\text{next}} + a_{\text{before}} = 2 \cos \omega a_{\text{now}}$, so $2 \cos \omega = \frac{\text{num}}{3 \text{ den}}(4 + 2 \cos k)$. $\square$

Proposition 2 (the pace is conformal). *At a Node whose pace is p on every axis, every family's rotation at every wave number obeys*

$$\sin^2 \frac{\omega}{2} = \left(\frac{p}{\Gamma}\right)^2 \sin^2 \frac{\omega_{\text{vac}}(k)}{2},$$

$\omega_{\text{vac}}(k)$ the rotation of Proposition 1: the pace multiplies the clock of every wave and every bound mode alike.

Proof. At an isotropic pace $S = 12 \text{ den } (\Gamma^2 - p^2)$ and $R_a = 2 \text{ num } p^2$, so $2 \cos \omega = \frac{\text{num } p^2}{3 \text{ den } \Gamma^2} (4 + 2 \cos k) + 2 \left(1 - \frac{p^2}{\Gamma^2}\right) = 2 - 2 \left(\frac{p}{\Gamma}\right)^2 \left(1 - \frac{\text{num}}{3 \text{ den}} (2 + \cos k)\right) = 2 - 2 \left(\frac{p}{\Gamma}\right)^2 (1 - \cos \omega_{\text{vac}})$, and $1 - \cos \omega = 2 \sin^2(\omega/2)$. $\square$

Two clocks at one level shift alike, $d \ln \omega / dc = 2 \tan(\omega/2) / (\omega p)$ per unit of the held level c , and a packet's fall in a pace gradient does not depend on its family or its shape beyond its band's own [1, the line]: this is the paper's equivalence principle, a reading of the line and no postulate. The clock's slowing is this pace: a record at a Node of pace p rotates and moves as a record at Γ does, with its intervals p/Γ as long.

2.3 The direction

Theorem 1 (the step is reversible bit for bit). *Let σ be the direction, +1 forward and -1 backward, div and mod the floor division and its remainder in $[0, w)$, and ρ the remainder carried in. With*

$$u = \sigma \left(\sum_a R_a \text{arr}_a + S x - w y \right) + \rho, \quad z = \sigma(u \text{ div } w), \quad \rho' = u \text{ mod } w,$$

forward $(x, y, \rho) = (a_{\text{now}}, a_{\text{before}}, r)$ gives $(z, \rho') = (a_{\text{next}}, r')$, and backward $(x, y, \rho) = (a_{\text{now}}, a_{\text{next}}, r')$ gives $(z, \rho') = (a_{\text{before}}, r)$.

Proof. Forward, $\sigma = 1$, the line is the definition of div and mod . Backward, $\sigma = -1$: write Y for the right side's first two terms, $\sum_a R_a \text{arr}_a + S a_{\text{now}}$. Then $u = w a_{\text{next}} - Y + r'$ and $z = -\lfloor u/w \rfloor = \lceil (Y - w a_{\text{next}} - r')/w \rceil$. From the forward line, $Y - w a_{\text{next}} - r' = w a_{\text{before}} - r$ with $0 \leq r < w$, whose ceiling divided by w is a_{before} exactly; and $\rho' = u - w \lfloor u/w \rfloor = w a_{\text{before}} - (Y - w a_{\text{next}} - r') = r$. So the backward call returns (a_{before}, r) bit for bit, and the reversibility of every run needs no second function [1, the direction]. $\square$

The whole interval is a bijection but for the click: the places of the interval (Section 2.9) run backward in reverse order, each primitive by its own inverse, and the count's line by the same line with the current reversed (Section 3.2).

2.4 The four acts and what they can form

Rule3's line is one act; declared coefficients make it four [1, the four acts]: (a) *the read*, the line over the GameBoard with the six arrivals; (b) *the one-Node step*, the line with the neighbours'

reads declared 0, $a_{\text{next}} + a_{\text{before}} = (a/b) a_{\text{now}}$, the rotation by the angle whose doubled cosine is a/b (Chebyshev's recurrence); with $a/b = 1$ and the coefficient on a_{before} declared 0 it keeps a level, and with $a/b = 2$ from $(0, 1)$ it counts, $a_t = t$; (c) *the division*, the line with $R = S = 0$, $w a_{\text{next}} + r' =$ the numerator $+ r$, Euclid's division with the remainder kept; (d) *the load*, a declared integer in the numerator, how a count enters a level. A composition of the acts is a finite sequence of them, each with its coefficients (a pair, a weight, a wall, a load) declared for the world, applied to declared levels.

Theorem 2 (what the acts can form). *Every composition of the four acts is a piecewise-linear function of the levels with integer slopes: a sum of linear forms and of floors of linear forms, composed. No such function is quadratic in the levels.*

Proof. Each act is a linear form of the levels or the floor of one; a floor of a linear form is piecewise linear with integer slopes; sums and compositions of piecewise-linear functions with integer slopes are of the same kind. The second difference of such a function in any one level is 0 almost everywhere, while the second difference of a product $\text{now}_i \text{before}_j$ in the pair $(\text{now}_i, \text{before}_j)$ is 1. $\square$

What the theorem separates: every product of two levels in the law is a *booking*, a reading of a bilinear form with declared coefficients and no write (Section 3.1); the one thing written from a product is the count, by the count's line, and that product is Rule3's own conserved current (Theorem 3). *The three tests of every rule:* a line enters the law only if it is generic (one primitive with declared integers, no family's name and no kind), vector (one of the acts on the state, no root and no float) and local (its own record and the six neighbours, nothing kept at a Node beyond the law's numbers). A hypothesis that needs more is stated under its own name, outside the law.

2.5 The paces and the guard

The *pace* of a Node for a family is what its reads make of the Node clock $[1, \text{the paces}]$:

$$\begin{aligned} p_0 &= \Gamma - \sum_{\text{reads}} (\text{weight} \times \text{by} \times \text{the read family's time level}), \\ p_a &= p_0 - \sum_{\text{reads}} (\text{the same}) - \sum_{\text{reads}} (\text{weight} \times \text{by} \times \text{its } aa \text{ component div } 2), \quad a = x, y, z, \end{aligned} \tag{4}$$

one division per read per axis, rounded at the read: the pace is a coefficient of the interval and no level, so no remainder is kept for it. A positive weight is a hollow (the read level slows the clock, an attraction); a negative weight is a hill; by “ q ” the weight is multiplied by the reading record's charge sign. A family with no reads steps at $p_0 = p_a = \Gamma$, the plain rule. The time level lowers the clock's pace p_0 once and each axis's pace p_a once more, so a wave's speed along a Link falls by the level twice (the Link as well as the clock, as nature's light past a body bends by twice the clock's share) while a mode's rest rotation, from S alone, falls by the level once.

Proposition 3 (the guard). *The step is stable at a Node exactly while $0 < p \leq P$ with the edge*

$$P = \text{isqrt}(2 \text{den } \Gamma^2 \text{ div } (\text{den} + \text{num})),$$

Γ for light and above Γ where $\text{num} < \text{den}$; every pace at or below Γ is inside it.

Proof. The mode at wave number π on every axis has $S_6(a_{\text{now}}) = -6 a_{\text{now}}$, so its factor is $(S - 6R)/w = 2 - 2(1 + \text{num}/\text{den})(p/\Gamma)^2$, which is $-2 \text{num}/\text{den}$ at $p = \Gamma$ and falls below -2 , the edge of Chebyshev's recurrence, once $p^2(\text{den} + \text{num})$ exceeds $2 \text{den } \Gamma^2$; there the record grows without bound. P is the largest integer pace at which $p^2(\text{den} + \text{num}) \leq 2 \text{den } \Gamma^2$ holds. Below 0 the pace is no clock. $\square$

A hill lessens a hollow and never exceeds it; a pace outside the guard refuses the run, naming the Node. *The band at a pace*, a reading of the line and no new rule: at a Node of pace p on every axis a record's rotation at wave number k along an axis is

$$\cos \omega = 1 - \left(\frac{p}{\Gamma}\right)^2 \left(1 - \frac{\text{num}}{3 \text{den}} (2 + \cos k)\right), \quad (5)$$

so the band of rotations a body's Nodes carry narrows with p^2 : a record whose rotation lies outside the band there is evanescent inside (for $[1, 1]$, $\cosh \kappa = (\Gamma/p)^2(1 - \cos k) - 1$ per Node) and is reflected whole, and one whose rotation lies inside is slowed to the band's group velocity. Light meets a total mirror exactly where $p < \Gamma \sin(k/2)$, that is where the held count c exceeds $\Gamma(1 - \sin(k/2))$ (0.2929Γ at the wavelength 4 Links), and a window otherwise, with the step's partial share at each face. A body is a balance of a hollow and a hill of different ranges: a hollow alone in three dimensions binds no record (it collapses into its own well or disperses); gravity is the one-signed exception (Section 5).

2.6 The bound

The rule's total at a Node whose levels and arrivals stand at the amplitude A is at most $6A|R| + A|S| + w(A + 1)$. Every world declares integers under which this total sits below $2^{63} - 1$, the width of the integers, or the declaration is refused naming the total $[1, \text{the bound}]$. The amplitude unit A is derived, never written: the largest amplitude at which every total of an interval stays inside the width, the least of three, Rule3's $2wA$ with $w = 6\Gamma^3$ (Section 6.2), the count's line's $6A^2$ and the transport's $3d_1d_0(A + 1)$ with the largest denominator d of the twist table. With $\Gamma = 10^4$ and $A = 2^{20}$ Rule3's total is near 3×10^{18} , one third of the room (COMPUTED).

2.7 The transport

A family with a pair (re, im) at each Node and a read with a twist turns its arrivals by the Link's angle $[1, \text{the transport}]$. On the Port along the axis a in the sense σ , for each such read with the factor f (the weight times the record's own rotation for the twist "own", else the declared twist; by q the charge sign),

$$k = \sigma \times f \times (V_a \text{ at this Node} + V_a \text{ arrived through the Port}),$$

V_a the read family's vector component along the axis; both ends form the same number with opposite signs, so the transport back is the inverse rotation. The universe's *twist table* gives, for the angle k in units of 1/unit radians, a Pythagorean triple (c, s, d) with $c^2 + s^2 = d^2$ exactly: $|k| = k_1F + k_0$ with F the fine table's length, the fine triple of k_0 and the coarse triple of k_1 composed, $(c_1c_0 - s_1s_0, s_1c_0 + c_1s_0, d_1d_0)$, with $(c, -s, d)$ for $k < 0$ and $(1, 0, 1)$ at $k = 0$; an angle beyond the table refuses the run naming the Port. The arriving pair is rotated to the nearest unit,

$$R_{\text{re}} = (2c \text{re} - 2s \text{im} + d) \text{div } 2d, \quad R_{\text{im}} = (2s \text{re} + 2c \text{im} + d) \text{div } 2d,$$

two read acts of Rule3 with the coefficients $(2c, -2s)$ and $(2s, 2c)$ on the two arrivals, the load d , the wall $2d$, the remainder not kept: a remainder carried across intervals is one to one only while d stands, and d changes with the angle every interval; the rounding is unbiased in the mean, and the inverse recomputes the same triple from the start's levels, exact. The record's own rotation is written once at the load from the record's pair (matter) or from its wavelength on light's dispersion (light), so no per-world number is declared; the charge's twist on a charged record is "own" by q , and no separate number exists.

The second level and the self-source. A phase-2 family runs the line on each of its two levels with the transported arrivals; with no twist the second level stays exactly zero when it starts zero. The self-source of a family with the unit P_2 above 0, $\Sigma_{\text{self}} = (\sum (a_j - a_i)^2) \text{ div } P_2$ over the six Links, is subtracted from the step's right side as $w \Sigma_{\text{self}}$: each difference a read act, its square a booking of the family's own levels, the division Rule3's; $P_2 = 0$ turns the line off [1, the second level].

2.8 The conserved form

Theorem 3 (the conserved form). *Let the paces stand. The read matrix $2 \text{ num } p_i^2$ is not symmetric where the paces differ, so the form carries the weight $1/p_i^2$ on the Node terms and the plain current on the Links:*

$$E = \sum_i \frac{w (\text{now}_i^2 + \text{before}_i^2) - S_i \text{ now}_i \text{ before}_i}{p_i^2} - 2 \text{ num } \sum_i \text{ now}_i S_6(\text{before})_i.$$

E is exact where the paces stand and changes by the work term where they move; with the rounding, each Node's remainder adds at most one unit of the level per interval to the walk of E . Its Link term, the plain current $\text{num}(\text{now}_i \text{ before}_j - \text{before}_i \text{ now}_j)$ through the Link ij , is the current of Rule3's own conservation law and the one product the click reads.

Proof. Without the rounding the line is $\mathbf{a}_{\text{next}} + \mathbf{a}_{\text{before}} = \mathbf{M} \mathbf{a}_{\text{now}}$ with $\mathbf{M}$ the matrix whose diagonal is S_i/w and whose entries on the six Links are R/w . Let $\mathbf{D}$ be the diagonal matrix of the weights $1/p_i^2$; then $\mathbf{D}\mathbf{M}$ is symmetric, since $R_a = 2 \text{ num } p_a^2$ at the reading Node and the weight $1/p_i^2$ cancels it on both sides of every Link. For a symmetric $\mathbf{K} = \mathbf{D}\mathbf{M}$ the quantity $Q(\mathbf{x}, \mathbf{y}) = \mathbf{x} \cdot \mathbf{D}\mathbf{x} + \mathbf{y} \cdot \mathbf{D}\mathbf{y} - \mathbf{x} \cdot \mathbf{K}\mathbf{y}$ satisfies $Q(\mathbf{a}_{\text{next}}, \mathbf{a}_{\text{now}}) = Q(\mathbf{a}_{\text{now}}, \mathbf{a}_{\text{before}})$: with $\mathbf{a}_{\text{next}} = \mathbf{M}\mathbf{a}_{\text{now}} - \mathbf{a}_{\text{before}}$, $\mathbf{a}_{\text{next}} \cdot \mathbf{D}\mathbf{a}_{\text{next}} - \mathbf{a}_{\text{next}} \cdot \mathbf{K}\mathbf{a}_{\text{now}} = (\mathbf{M}\mathbf{a}_{\text{now}} - \mathbf{a}_{\text{before}}) \cdot \mathbf{D}(\mathbf{M}\mathbf{a}_{\text{now}} - \mathbf{a}_{\text{before}}) - (\mathbf{M}\mathbf{a}_{\text{now}} - \mathbf{a}_{\text{before}}) \cdot \mathbf{D}\mathbf{M}\mathbf{a}_{\text{now}} = \mathbf{a}_{\text{before}} \cdot \mathbf{D}\mathbf{a}_{\text{before}} - \mathbf{a}_{\text{before}} \cdot \mathbf{D}\mathbf{M}\mathbf{a}_{\text{now}}$, which is the other side. Multiplied by w this is E up to the Link term's sign convention [1, the conserved form]. $\square$

E is a GameBoard reading. Its Node term at a Node, $D_i = \text{now}_i^2 - \text{next}_i \times \text{before}_i$, is the record's form there, and the record's quanta at the Node are $D_i \text{ div } T$: the count is the record's form, and this is how a body's well is laid (Section 4).

2.9 The interval

The law steps every family from the state at the interval's start to the state at its end. Every read is of the start's values: the six neighbours' levels, the Node's own pace, the arrivals, never a value written in the same interval; one Link per interval and nothing in zero time. The interval has five places, in order [1, the interval]: (i) the clicking families' step, every component, with the transport through the Ports and the self-source; (ii) the bookings at the Ports (the current), the ladder, the takings and the givings, the count's line; (iii) the held families' step; (iv) the holds written (a body's values into its family's levels), the clicks' changes of a body's content, charge and momentum included; (v) the bodies on one Node: the feed, the induction, the spin's step, each a reading of the body's record and no write. The order within a place, where two primitives write one value, is declared once for every world. A click's writes enter at the next interval. Backward, the places run in reverse order and each primitive runs its own inverse.

3 The click

The click is the law's one non-local act and the only thing the Outside reads. This section states the booking, the count's line whose division is the click, the ladder that reads it today, and the theorems: the count is conserved, one quantum makes one click, Born's rule, and for a pair record the correlation $\cos 2(a - b)$ with no signalling [1, the click].

3.1 The booking

The *current* of a record into Node i from its neighbour j through their Link over an interval is the bilinear form of its two levels at the two ends,

$$F_{ij} = \text{weight} \times (\text{now}_i \text{ before}_j - \text{before}_i \text{ now}_j), \quad (6)$$

the pair's second level added, the weight the family's common wall in the form's units; positive inward. A record of wave number k travelling from i to j has $F_{ij} = -2 \times \text{weight} \times |\psi|^2 \sin \omega \sin k$, below 0. It is antisymmetric, $F_{ji} = -F_{ij}$, and it is the Link term of the conserved form of Theorem 3: the current of Rule3's own conservation law. The booking reads it at a Port and writes nothing.

3.2 The count's line

The *family of clicks* is one family like every other: its level at a Node is the *count* c , the number of quanta there; its wall is T , the quantum's action; its read is the current. At every Node,

$$T c_{\text{next}} + r' = T c_{\text{now}} + \sum_j F_{ij} + r, \quad 0 \leq r' < T, \quad (7)$$

the sum over the six Ports of the current into the Node over the interval, r the remainder kept at the Node: Rule3 with the coefficient 1 on each axis's inflow through its two Ports, T on the count and the wall T . The *click* is the division: a quantum moves whole from one Node to the next exactly when the remainder crosses T , and never a fraction of one. The record ends where its last quantum leaves it; the count that arrives at a detector's Node is the detector's click, and a detector's count is the family of clicks' level at its Nodes; an emitter's giving is the count leaving through its outer Ports into the born record. The same line carries a body: the count follows its record's current at every Node, one quantum at a time, so a body moves without any accumulator of its position. The remainder at a Node starts at the ladder's origin, $(2u + 1)T \text{ div } 2$ with u the record's residue (Section 3.3); a body's count is laid with its record, at every Node $Tc + r$ the Node's share of the record's form plus the origin, so the count follows the norm with the margin $T/2$ against the rounding's walk. The inverse is the same line with the current reversed, exact.

Theorem 4 (the count is conserved). *Over any region of Nodes, $\sum (Tc_i + r_i)$ changes in one interval only by the currents through the region's boundary Links.*

Proof. Summing Eq. (7) over the region, each interior Link's current enters one Node's right side as $+F$ and its neighbour's as $-F$ and cancels; what remains is the sum over the boundary Links, a telescoping sum, exact in integers. On a periodic GameBoard with no face the sum is constant to the bit. $\square$

The velocity. The record's wave number is conserved by the record's line, and the count's line is the continuity equation of the current in integers, so the count's centroid moves at the current's velocity, with no leak beyond the ladder's remainder; a body at rest has zero net current at every

Node in the mean and its count stays in place. A record of two levels with the phase k per Link carries its count at Rule3's group velocity, $R \sin k / (w \sin \omega)$ (Section 5.4). Under the count's line a moving body is the moving mode of the generator, its count moved by its record's current.

3.3 The ladder

At a detector the click is read by the *ladder*, the same division act on a record's one-way inward flux [1, the ladder]. The data: the record's residue u (its remainder at its giving Node), its norm T , its wheel V , hence its threshold $\theta = (2u + 1)T/(2V)$, fixed at the giving; the named detectors in their declared order with the face last. At every interval each Node i of a detector receives the one-way inward flux $f_i(t) \geq 0$ of the record through its Ports (a Link between two Nodes of one detector is not a Port and carries no offer); the record's running total is $C(t) = C(t-1) + \sum_i f_i(t)$. The click is at the first interval t^* with $C(t^*) \geq \theta$, at the Node whose segment of that interval's increment, laid in the ladder's order, contains $\theta - C(t^* - 1)$; there the record ends, whole. The rungs are the division act, $b_k = (2VC_k + \text{Total}) \text{div } (2 \text{Total})$.

Theorem 5 (one quantum, one click). *θ is crossed exactly once.*

Proof. C is non-decreasing, $\theta < T$, and the total offered with the face last is at least T (what leaves offers everything at the face), so C crosses θ once; the deletion of the record at once leaves nothing to cross it again. $\square$

Theorem 6 (Born's rule). *Let u be spread evenly over $\{0, \dots, V-1\}$; then the probability that the click lands at Node i is $C_i(\infty)/T$, the detector's share of the record's total inward flux, whatever the time profile.*

Proof. θ is then spread evenly over V points in $[0, T)$. The segments $[C(t-1) + \sum_{j < i} f_j(t), C(t-1) + \sum_{j \leq i} f_j(t))$, one per (interval, Node), partition $[0, C(\infty))$ and have the lengths $f_i(t)$; so the probability of the click at interval t at Node i is $f_i(t)/T$ to the grain $1/V$, and summing over t gives the share. The counts over many records are binomial about the shares, since the residues are equidistributed and not a permutation. $\square$

The residue is the rule's own remainder at the giving Node, never declared and never drawn: chance lives in the remainders, which stay at the Nodes, bounded, never reset and never erased. A detector is one connected region of Nodes with one name; separate places are separate names. A detector *set* is one Node of the count's line with its Ports the set's outer Ports, the sum of its Nodes' lines with one remainder; Born's rule is the shares of the current among the set's Nodes.

3.4 The pair record and Bell's correlation

The pair record, rank 2. One record with two labels, its level at a Node the 2×2 matrix $\mathbf{M}$ over the labels' two directions, written as the identity by a crystal's giving click (the two labels identical), stepped by Rule3 alone at every Node it reaches (the step, the transport and the paces as any record's, component by component, Rule3 linear), spreading from the crystal's Node through the Ports to both sides; a polariser turns the labels at its own Nodes by its angle through the transport's exact triple, the rows on the one side and the columns on the other, reading nothing beyond them [1, the ladder, the pair record]. The crystal is a body of matter with one key and one Node: its taking click takes the arriving quantum whole into its count, and in the same interval its giving click lowers its count by one and gives the pair record at $T \text{div } 2$ by the division act, each label one quantum of the record's own unit, the half; the theorem of one quantum per click holds

per label in the label's unit, and the pair's two clicks are the arriving quantum, a half and a half, as nature's down-conversion.

The labels' clicks. Each label clicks alone at its own detector, by its own ladder (its own residue, the two spread at the crystal's Node), at the interval the label reaches the detector's Nodes, reading nothing but the level there. The detector of the first label at the setting a offers its two sets the rows of $\mathbf{R}(-a)\mathbf{M}$, each row's booking one half, and the taking click of the row i ends that label and not the pair: the record goes on as rank 1, its level at every Node the row i of the turned matrix there (the click, the one non-local act, ends one label at once everywhere and leaves the other on the GameBoard), so the second label at the setting b offers $\mathbf{R}(-b)$ applied to that row.

Proposition 4 (the pair's shares). *With $\mathbf{M}$ the identity at the giving, the first click lands on either of its side's two sets with the share $1/2$ at every setting; the second click lands on the set of the same index with the share $\cos^2(a - b)$ and on the other with $\sin^2(a - b)$, whichever label clicks first. So over all pairs*

$$E(a, b) = \cos^2(a - b) - \sin^2(a - b) = \cos 2(a - b),$$

each side's marginal is $1/2$ at every setting of the other side (no signalling), and the Clauser-Horne-Shimony-Holt sum [3] at the settings $0, \pi/4$ for one side and $\pi/8, 3\pi/8$ for the other is $S = 2\sqrt{2} = 2.83$, Tsirelson's bound [4], derived from the local law with the click's one non-local act alone.

Proof. The rows of the rotation $\mathbf{R}(-a)$ are two unit vectors, so each row's booking is one half of the label's norm: the first click's shares are $1/2$ and $1/2$. After the click of the row i the record's level is that row, the unit vector at the angle $-a + i\pi/2$; the second polariser applies $\mathbf{R}(-b)$, whose components on the two sets are $\cos(a - b)$ and $\pm\sin(a - b)$ up to the index's quarter turn, so the shares are their squares. E is the difference of the shares of the same and the other index. Summed over the first click's index at $1/2$ each, the second side's marginal is $\frac{1}{2}(\cos^2 + \sin^2) = \frac{1}{2}$, whatever a ; the first side's is $1/2$ by the first step, whatever b . The sum $S = E(a, b) - E(a, b') + E(a', b) + E(a', b')$ at $a = 0, a' = \pi/4, b = \pi/8, b' = 3\pi/8$ has $E(0, \pi/8) = E(\pi/4, \pi/8) = E(\pi/4, 3\pi/8) = \cos(\pi/4)$ and $E(0, 3\pi/8) = \cos(3\pi/4) = -\cos(\pi/4)$, which enters with the minus sign, so $S = 4\cos(\pi/4) = 2\sqrt{2}$. $\square$

The correlation travels in the record alone, by Rule3, one Link per interval; the click ends one label everywhere at once and carries no information, since each side's marginal does not depend on the other's setting: the law is local in its transport and non-local in the click only, as Bell's theorem requires of any account that reaches $2\sqrt{2}$ [2]. The pairs are the host's coincidence reading, not law: the two clicks' intervals within a window declared beside the world, as Aspect's [5]; the law has no coincidence set and no joint click, since no uniform time of events exists on the GameBoard. With switching polarisers, each side's angle is the one in force at the label's passage, by the polariser's own clock, the count act on the body. A click on a record with a partner label is entanglement and a click on a record with none is a measurement, and Bell's S reads that and nothing else [1, the universe is bound].

4 The one write and the primitives

A family's write is one act. Every write into a family's level on the GameBoard is $(w \times q(\text{Node}) + r) \div E$ at a Node each interval, at both levels of the writer's rotation (the second the read act once more, halved), the division act with the remainder r carried at that Node [1, the

primitives]. Here $q(\text{Node})$ is the writer's own quantity there: for a body, the count declared there times its tensor $(1, n_a/W, n_a n_b/W^2)$ with $\mathbf{n}$ its momentum and W its wall; for a record, its quanta there, $D_i \text{ div } T$, times its rotation over the rest rotation of the matter quantum, ω_r/ω_m (the energy sources, as a photon weighs $h\nu/c^2$); for the giving, the body's record's two levels at its shell times the bound charge it holds there; for the recoil, the click's tally. w is the one weight with which the writer's family reads the written family (whoever reads with w sources with w : action and reaction one key), and E the written row's divisor. The five instances are the hold, the source, the giving, the start (the rest of the hold's line, its fixed point) and the recoil, and no number is the act's own: generic (the rows' keys), vector (one carried division), local (the Node). Under the divisor 1 the level is the count itself, written once and never added interval after interval: the count is read once (Section 7).

The hold. A body of quanta M_k of the family k has the count $s = \sum_k w_k M_k$, the charge Q , the momentum $\mathbf{n} = \mathbf{j} \text{ div } T$ with $\mathbf{j}$ the count's line's current through its Ports (a reading of its record; its velocity $\mathbf{n}/M$ in Links per interval), the spin $\mathbf{S}$ the curl of that current about its centre and the moment the same with the charge's sign. At every Node of its support it writes into the families it holds: gravity's time part the count declared there, over the row's divisor E_s , with a remainder of its own at that Node, never the whole s at every Node, so that two bodies' wells add and the field's rest is Poisson's with the sources; its vector part $j_a \text{ div } (E_s T)$ and its tensor part $(n_a n_b) \text{ div } (E_s M)$; the charge's time part the count times the body's sign q (a family carries no sign; a body's key signs its count, so one matter family holds bodies of both signs) and its vector the same current with the sign; and the dipoles at the Node's six neighbours. The three parts are one source, the count and its current's tensor at one scale with no factor declared: the linearised field's 4 and 2 were nature's numbers by hand and left. At a body with a record the count written is its own record's quanta there, $D_i \text{ div } T$ with the form's remainder carried, so the well moves with the record to the form's resolution and the self-force vanishes.

The source. A record writes as a body does: at its Nodes the sourced family's level gains $w \times ((D_i \text{ div } T) \times (\omega_r/\omega_m) + r_i) \text{ div } E_s$ each interval, with the same w the record's family reads the sourced family by; so light sources gravity and the bound charge as its quanta do ($E = mc^2$ on the comparison side), and every pair of families that read each other is reciprocal. The field's shape is the static limit of Rule3, $(\Delta - \kappa^2) a = -\sigma/q$ with $\kappa^2 = 6 \text{ den/num} - 6$, κ the inverse reach, σ the source and q the weight.

The giving. A body that gives writes the given family's level at its outer shell, the Nodes with a Port to a Node outside the body: each interval the shell gains $(M_{\text{pol}} \times a_{\text{body}}) \text{ div } E_s$ at both levels, the family's write by the body's record with the bound charge M_{pol} it holds there, so a neutral atom shines, the flux grows as M_{pol}^2 , and the given row's divisor is the one coupling of light to charge, as e in nature. The window's three acts are the open (the given family's count at the body down by one, the count's line's move with the opposite sign), the write, and the close at an outward flux of T , the universe's quantum action, at which the record is named with its direction the tally's sign per axis; one quantum is the train the window writes, N_q periods of the giver's rotation until the outward flux reaches T . One quantum per window bounds the charge's divisor from below (Section 7).

The recoil. At the close the body's record turns by $\delta k = \sigma_a \times (k_q \text{ div } M)$ per Link along each axis with a tally, the rotation act by the table's triple nearest δk , the angle's remainder carried at the body, the taker and the giver with opposite signs; so the record's envelope carries the phase k per Link of a moving body and the velocity moves by the quantum's momentum over the body's

quanta, a heavy body's recoil small by its quanta. The momentum $\mathbf{n}$ is read from the record's current and is no level; a taker with no record of its own takes no recoil.

What is derived and no primitive. A body has no law of motion of its own. Its record moves by Rule3 alone, and in a pace gradient its packet accelerates at $a = \Delta \times |d\omega_0/dc| \times (c_+ - c_-)/D$, Δ the mode's top velocity, $d\omega_0/dc$ its redshift per level, c_+ and c_- the read level at its two faces D Links apart; the count follows the record by the count's line; bodies with the same $\Delta \times d\omega_0/dc$ fall alike. The reader's pace carries the read family's vector part, $-(n_b V_b) \text{ div } W$, so the record's packet turns and drifts in a vector part's gradient by Rule3 alone: the Lorentz force and its gravitational twin with no line of their own. The spin turns with the record by Rule3 alone; the precession of a spinning body is a reading of its record. Every primitive of the law is one line of its table [1, the primitives], each an act of Rule3 with declared coefficients and nothing else.

5 The stable body

5.1 What a body is

A body is its count and its record [1, the stable body]. The count c_i is the family of clicks' level at the body's Nodes, in quanta; it enters Rule3 through the pace alone, $p_i = \Gamma -$ the held level at the Node, and it never splits: a quantum moves whole, by the count's line. The record (now, before) of the body's family splits by the send and is bound by the count. A body's declaration names its family, its Nodes, its count per Node, its momentum at its two levels and its spin's two levels where it has one, and on a body that gives its stocks and its emitter, and nothing else: no lowered pair on its Nodes, no seed, no stop, no profile, no closed Port, no period.

5.2 The well and the reach

At a Node whose held level is c , Rule3's one-Node rotation rises from $\cos \omega_0 = \text{num}/\text{den}$ by $(1 - p^2/\Gamma^2)(\text{den} - \text{num})/\text{den}$, and its bonds fall from $\text{num}/(3 \text{den})$ to $\text{num} p_i p_j / (3 \text{den} \Gamma^2)$: the symmetric form of the line with the weights $1/p_i$. The depth is bounded by the family's own $1 - \cos \omega_0$, so a family whose pair lies close to 1 binds nothing. A body's clock is set by the level at its Nodes and its side, from the rule: a body's mass is its count.

The reach of a held family is its pair. A held family's rest solves $\text{num} \Delta^2 a - 6(\text{den} - \text{num}) a = -3 \text{den} \sigma$, Δ^2 the six-neighbour second difference and σ the source once an interval, so outside a body its rest falls by $e^{-\kappa}$ per Link with

$$\cosh \kappa = \frac{3 \text{den}}{\text{num}} - 2, \quad (8)$$

the family's own band at zero rotation (the evanescent k of Proposition 1 at $\omega = 0$; $\kappa^2 = 6 \text{den}/\text{num} - 6$ to the second order), the same pair that sets the band of its records and de Broglie's wave number for them: the gap is the mass of the records and the inverse reach of the field, one number. The pair [1, 1] has no gap and its rest is Poisson's over the whole GameBoard, the pull between bodies; a pair with $\text{den} > \text{num}$ has the gap $2 \cos \omega = 2 \text{num}/\text{den}$ at $k = 0$, so at [1, 2] with the divisor 1 the level inside a body is its count and falls by $e^{-\kappa} = 0.127$ per Link outside ($\cosh \kappa = 4$). A body's well is the static field of every family it holds, read at its Nodes by the paces: a body's binding, a mirror's line and a window are the short-reach family's, and the pull between bodies the massless one's, two held rows of the universe's declaration, one pair each.

5.3 The generator

The generator builds every world from zero; it declares nothing and holds no float [1, the generator]. (a) Its input is a world in the law's form: the GameBoard, the Node clock, the universe it names and one body by its Nodes with their counts; the iteration runs on the body's Nodes and their surroundings, out to where the mode's tail falls below one unit. (b) The iteration is Rule3's read act with the before-coefficient 0 on the region, then the division act to the amplitude unit A : the power iteration of the symmetric form's top mode, the bound mode above the band. (c) The profile is an integer vector bounded by A , so the iteration is a map on a finite set and enters a cycle; the stop is the first repeat, exact, no tolerance and no declared count. (d) The mode's second level is the read act once more, halved; the two levels are scaled together so that the record's conserved form equals cT , the body's count times the quantum's action: the body's record carries exactly its quanta's action. (e) The moving body is the resting envelope with the phase k per Link along the axis of motion, the rotation act on its two levels by the triple of the pair (m, j) , $\cos k = (m^2 - j^2)/(m^2 + j^2)$, exact; k is the pair at which the packet's own current along the axis at the norm T equals the count's line's velocity of the packet, by bisection on j at the declared m . (f) The amplitude unit A is derived, never written: the bound of Section 2.6 solved for the amplitude; the fixed point does not depend on A beyond its resolution. (g) The field at rest, the start of a world: the held family's own line is iterated to rest first, the count over the divisor added at the body's Nodes each pass, the homogeneous line outside; the map is monotone, so the levels rise to a fixed point, Poisson's rest with the sources within one unit; a world starts from the field at rest, never from zeros, whose transient rings. The start writes every held family at its rest once before the first interval, the level and its remainder together: the fixed point of the division act is a pair, and a remainder 0 by fiat is no rest.

In the rule's own universe the generator is Rule3 itself (Section 7): a body's bound record is what Rule3 makes of its count at its Node in the rule's own integers, run until the record stands, and its period, amplitude, clock pair, tail and wavelength are readings of the standing record and never inputs.

5.4 The velocity

Rule3 is translation-invariant, so the record's wave number k is conserved where no pace gradient stands: the record's centre moves at its group velocity $v_g(k)$ exactly, bound or spreading. The count follows the record by the count's line: with the record's norm cT the current across the body's face is cTv_g per interval, so cv_g quanta cross per interval and the count's centroid moves at v_g , exact up to the remainder [1, the velocity]. The well moves with the count one quantum at a time, $1/c_i$ of a Link, so the record is never kicked by a whole Link. A one-Node body of one quantum binds nothing and is a free record: its velocity is $v_g(k)$ with no count to follow it. A larger fall of the bound norm, or a slope that drifts, names a missing law; the body's own numbers are never patched.

6 The families from the rule

A family is a band of the one rule and not a declaration, and its row holds nothing the rule and the geometry fix [1, the families from the rule]. This section derives the exact bands from the one-Node line, writes every pair over the one denominator, sorts the families by the three tensor ranks of the three axes, and states under their own names the hypotheses on the quarks, the weak interaction and the electron.

6.1 The exact bands

Proposition 5 (the three exact rotations). *The line of one Node at rest, $a_{\text{next}} = 2 \cos \omega_0 a_{\text{now}} - a_{\text{before}}$, is exact in integers, leaving no remainder from any integer start, exactly where $2 \cos \omega_0$ is an integer. The rotations among them are $2 \cos \omega_0 = 1, 0$ and -1 , the periods 6, 4 and 3, the pairs $[1, 2]$, $[0, 1]$ and $[-1, 2]$; $2 \cos \omega_0 = 2$ keeps a level and -2 alternates it, no rotation.*

Proof. With integer levels the right side is an integer exactly when $2 \cos \omega_0 a_{\text{now}}$ is, for every a_{now} , that is when $2 \cos \omega_0 \in \mathbb{Z}$; $|2 \cos \omega_0| \leq 2$ leaves the five values, of which ± 2 are the degenerate recurrences. The period of $a_{\text{next}} = a_{\text{now}} - a_{\text{before}}$ is 6, of $a_{\text{next}} = -a_{\text{before}}$ is 4 and of $a_{\text{next}} = -a_{\text{now}} - a_{\text{before}}$ is 3. $\square$

A record on an exact band at rest at the vacuum's pace leaves no remainder in its rest rotation, so its quanta stand exactly at every Node, it gives nothing and never decays alone: stable matter is built on an exact band. Everything else of it clicks as every family's does: its packets (a wave number is no integer rotation), its Nodes in a well and its reads, so an exact band is seen exactly when it is moved or probed and only then. *The exact bands are exact only in the vacuum:* at a Node of pace p_0 the band's rest rotation is, by Proposition 2 at $k = 0$,

$$2 \cos \omega = 2 - 2 \left(\frac{p_0}{\Gamma} \right)^2 \frac{\text{den} - \text{num}}{\text{den}},$$

an integer at $p_0 = \Gamma$ alone for the three pairs, so a record on an exact band inside a well, its own or another's, leaves a remainder and clicks like every family's; the three phases of the third (Section 6.4) sum not to 0 but to a share of a record's level that grows with the well, 0.015 at a level 30 of 12,000, 0.148 at 300 and 1.2 at the self-binding edge (COMPUTED), and white matter stands only where the well at its Nodes is small against Γ . Under a divisor far above 1 a white body of thousands of quanta is white to 10^{-5} ; so white matter requires a divisor far above 1, and the smallness of the charge's coupling is a condition of the quarks' whiteness before it is a distance of nature from the rule.

6.2 The mass is one integer; the wall is one

Every pair is written over the GameBoard's one denominator Γ , $\cos \omega_0 = m/\Gamma$, the declaration holding the numerator m alone: light $m = \Gamma$, the exact band $m = \Gamma \text{ div } 2$ (Γ even), matter its one integer, 6,667 in the universe of record (the nearest to $2/3$, every number written before a run within 5×10^{-5} of itself). The wall is $w = 6\Gamma^3$ for every family; the pair enters the coefficients as the file writes it, m over Γ ($R_a = 2m p_a^2$, the rest term $12(\Gamma - m)p_0^2$), and a reduction to lowest terms is a reading and never the loader's act, so every family's remainder has the one resolution and the exact bands leave none in any form ($S + 6R = 6\Gamma^3$ exactly at the vacuum's pace for $m = \Gamma \text{ div } 2$). With the one wall the amplitude unit of the width falls to $A = 512,409$ in the universe of record (COMPUTED), and T is set once so that every record at cT stays under A .

6.3 The final unification

Proposition 6 (the three reads of the line). *Of a held family, Rule3 reads three kinds of component and no fourth: its time level, into the clock's pace p_0 ; its vector component V_a , into the Link's angle; and its aa components, into the axis's pace p_a . These are a scalar, a vector and the diagonal of a symmetric tensor of rank two over the three axes, with 1, 3 and 6 components under the 48 signed permutations; a component of rank three enters no coefficient of the line, so a family of rank three would be read by no family and is no family.*

Proof. By Eq. (4) and the transport's angle (Section 2.7), the coefficients w , R_a and S and the angle k are functions of Γ , the pair, and the time levels, the V_a and the aa components read at the Node and through the Port, and of nothing else. The symmetric tensors over three axes of the ranks 0, 1 and 2 have 1, 3 and 6 independent components. $\square$

So the three of Rule3 is the three axes, and the three reads are the three ranks of what a family writes: the count, the current and the current's tensor (the hold, Section 4), the writer's side of the same three. Every family is one rank and carries two keys, its pair and its divisor.

Proposition 7 (the gap and the reach). *A family's band reaches $\cos \omega = 1$ exactly when $\text{num} = \text{den}$; a held family with $\text{den} > \text{num}$ has the gap $2 \cos \omega_0 = 2 \text{num}/\text{den}$ and the finite reach $1/\kappa$ of Eq. (8), and the held family of unbounded reach is $[1, 1]$ alone, whose rest is Poisson's.*

Proof. By Proposition 1 the band's top is num/den , which is 1 exactly when $\text{num} = \text{den}$; and $\cosh \kappa = 3 \text{den}/\text{num} - 2$ equals 1 exactly then, while for $\text{den} > \text{num}$ it exceeds 1 and the rest falls by $e^{-\kappa}$ per Link (Section 5.2). $\square$

One free number per rank (DERIVED). ASSUMPTION, named: the fields that every family reads into its pace at every distance, the pull between bodies and the charge's field, are fields of unbounded reach, and a quantized scalar holds nothing. Then a field every family reads is on $[1, 1]$ (Proposition 7), and its divisor is free, since Rule3 is linear and its form's scale is free: one count writes one level over E_s at any E_s , and no line of the rule fixes it; the vector's divisor is the fine-structure constant α and the tensor's is Newton's G (Section 8.2). A quantized scalar holds nothing, has no divisor, and its pair is free: the mass m . An exact band has both fixed, the pair by Proposition 5 and the divisor 1 by the well being the count, so it carries none. Thus each rank carries one free number and no more, the scalar the mass, the vector α , the tensor G : the count three is derived from the three reads, the values free as nature's. The crack, named: a second massive scalar family, a second mass at rank 1, the rule does not forbid; the universe of record declares one, and nature's spectrum of masses is the open row under this line.

A family's parts are the tensor rank of what writes it (1 the count; $[1, 3]$ the count and the current; $[1, 3, 6]$ the count, the current and the current's tensor); its phase is 2 where it carries quanta (a current, the count's line, the clicks) and 1 where it never clicks, the highest rank, which every family writes and which reads none, and the exact band, whose rotation leaves no remainder: two real fields. Its reads are the graph of the ranks, each family reading the ranks above it, with the one weight 1 (reciprocity), the reader's own twist and by its own count with the sign of the body that holds it; its held writer writes the count with the body's sign and the dipole of the body's spin with the divisor equal to its phase (1 for the real field, 2 for a pair, Dirac's 2). The action of one quantum is T , the universe's one integer for every family: Rule3 is linear and its form's scale is free, so no line of Rule3 fixes T ; it is a unit, declared once. So the declarations hold Γ , T , the matter pair and the two divisors, and nothing else of physics.

6.4 The third and the conversion

Hypothesis 1 (the third: the quarks and the strong force). A record on $[-1, 2]$ turns a third per interval; three thirds in the phases 0, $2\pi/3$ and $4\pi/3$, three consecutive states of one record, sum to zero in their linear write at every interval while their quanta add and stand constant (the white body), and the whiteness is read at every Node from its own three levels, the click's local language. When a third moves off its body every Node between the two carries the three levels' sum uncanceled, a record of the third's own count per Node, so the tube between them costs the third's

count per Link and its cost is linear in the length: the tension is the count, no number, the strong force's line; when the tube's quanta reach a third's worth the click converts them into a new pair of thirds, the string breaks into mesons and a third is never free. The three colours are the three phases; a third's charge reads a third, the sense of a third is the sign q of its read, and a meson is two thirds of opposite sign.

Hypothesis 2 (the conversion: the weak force). A click that takes a quantum of one family and gives it as quanta of others, the rotations adding as the crystal's row has it, is the whole weak interaction: one Node's act, no field and no massive vector, and no family's row declares anything for it.

A pair's numerator may be negative: the third's $R_a = 2 \text{num } p_a^2$ is negative, S is what the line makes it, the band is the mirror of $[1, 2]$ ($2 \cos \omega$ inside $[-1, 1]$ at every wave number and every pace at or below Γ , the edge $P = 2\Gamma$ by Proposition 3, the rest term $36p_0^2$), and a pair with $\text{den} > |\text{num}|$ is admitted as a massive kind; nothing of Rule3 changes. The three colours are three interval states: a third's record at rest steps (now, before) $\rightarrow$ ($-\text{now} - \text{before}$, now) with the period 3, so its three phases are three bodies whose two levels are three consecutive states of one record (7, -10 , 3 on one Node: each interval's three levels sum to 0 exactly, the quanta $7^2 + 10^2 + 3^2 - 7(-10) - (-10)3 - 3 \times 7 = 177$ constant), and the generator writes two levels and no new key.

A third does not bind itself: its read is negative, so its own well pushes its rest rotation up into the vacuum's band ($2 \cos \omega = 0.31$ at a count 3,000 of 12,000, inside $[-1, 1]$, COMPUTATION) and the record leaks; a record of the third at one Node is a packet of every wave number and disperses in every universe, its uniform mode alone exact. So one family on $[-1, 2]$ holds no standing body, no white body and no tube, and the white body of Hypothesis 1 stands only with the colours as three families of one pair under one constraint of whiteness, or, the line taken in the cluster universe of the divisor 1 (Section 7.5): *the colours are the three axes*. The white body is a one-Node body of $[2, 3]$; the count's line carries a tally per axis, so the three colours are the three tallies $\sigma_x, \sigma_y, \sigma_z$, and whiteness is their balance at rest (an isotropic tail carries no net current through any Port); under a pull the clicks are biased along one axis at the rate $\omega_b e^{-\kappa d}$ per interval, which falls with the distance and gives no linear potential: the force between two pixels is short-ranged, its range $1/\kappa$ (2.2 Links at 3,000 of 12,000; the click's reach 18 Links at $T = 1$), the nuclear force between nucleons and no confinement inside a one-Node body, which has no inside; the smallest bound fraction of a horizon body is $0.2255/(1/2) = 0.451$, so a third of such a body never binds and disperses (confinement is the edge, no line of its own) while two thirds bind, and a meson is two fractions above 0.451 of opposite q within the tail's reach. The charges $2/3$ and $1/3$ as numbers stay open (Section 9).

6.5 The electron is the lightest band

Hypothesis 3 (the electron). The lightest massive band the rule holds is $m = \Gamma - 1$, $\cos \omega_e = 1 - 1/\Gamma$, $\omega_e = \sqrt{2/\Gamma}$, and the electron is one quantum on it with $q = -1$: no bound body (in the cluster universe a bound body's ratio of counts is at most 2.22, Section 7.5), stable by the integer (one quantum does not disperse into fractions and no massive band lies below), its charge the proton's exactly since q is the click's integer; the positron is the same band with $q = +1$, and pair creation is the conversion.

The mass is the count: the rule reads inertia as the count, the momentum $n = Mv$ in the unit Q with the wall $W = 3QM$, so $m_p/m_e = c_p/1$, the proton, in the cluster universe, a one-Node body of 1,836 quanta, bound and within the horizon when $0.2255\Gamma \leq 1,836 \leq \Gamma \text{ div } 2$, so Γ lies between

3,672 (the horizon pixel) and 8,142 (the edge pixel), and the world's one number is measured; with the stable one-Node body being one, the smaller giving to the larger until it melts, the proton is the horizon body, $m_p/m_e = \Gamma \div 2$ and the atomic level's Γ is $3,672 = 6 \times 612$, one number and no range, a derivation under that hypothesis and no prediction of 1,836, which is its input; the universe of whole bodies leaves m_p/m_e open. What this carries by name: one quantum is one energy and its rotation is its frequency, nature's $E = h\nu$ being the record's click rate; the spin 1/2 is open.

7 The universe of whole bodies

The owner's word. No body is reduced to one Node. A body is the fixed point of its binding row on its Nodes; the form feeds fields with a range; the count moves by the count's line alone; the engine, the generator and the world stay generic, with no number in the engine. This section states the universe on those lines as its declaration now stands, the generator that makes a body, what the algebra says of a body's stability, and the hypothesis of the rule's own universe that came before it, with what became of it.

7.1 The declaration

The universe of whole bodies (DECLARATION, the files beside the law [1]): the Node clock Γ , 24 for the smallest checks and 6,000 for the looks; the quantum's action $T = 64$, one unit for every family; a *binding* family, the pair [23, 24] at $\Gamma = 24$ and [5,760, 6,000] at 6,000, a gapped pair whose reach by Eq. (8) is $1/\kappa$ of about two Links ($\cosh \kappa = 3 \times 24/23 - 2 = 1.130$, $\kappa = 0.51$), held with the divisor 2: the body's own well, with a flat bottom; a *gravity* family, the massless pair [1, 1], Poisson's rest and Rule3's steps, held with the divisor 10: the pull between bodies; the charge and the matter families; and no held row at the divisor 1, so the start solves every rest and the count moves by the count's line alone (Section 3.2), never laid from the form. Against the universe of record the free numbers are the two divisors and the binding pair; the hypothesis of zero free numbers (Section 7.5) is not what the files hold.

7.2 The generator makes a body

A body declared by its quanta M is laid over its Nodes as the fixed point of its binding row: the counts source every held row of the universe at the row's divisor by the start's own solver (Section 5.3); the body's standing record is what Rule3 makes in those paces, seeded by the integer power iteration within the body's region and scaled so that its form over the region carries M ; the counts are the form $D \div T$ with the remainder carried from Node to Node; each round takes the half step from the counts toward the form; the region is the Nodes carrying a quantum and one Link around them. Refused by name: a universe without T ; a cloud, the rotation not above the band's top; a collapse, a pace reaching 0; two bodies whose regions share a Node. The world is written with the body's Nodes and counts, and nothing of the body is declared beyond its quanta. Its shape is the rule's own:

Proposition 8 (the shape of the bound record). *Let the well be invariant under the 48 signed permutations about its centre (one Node, or a symmetric set), with the pair [num, den] and every pace inside the guard. Then (i) the bound mode, the top eigenvector of Rule3's operator, is unique, positive at every Node and invariant under all 48 maps; (ii) outside the well its decay in the direction $\mathbf{n}$ is the support function $h(\mathbf{n}) = \min_{\mathbf{q} \in \Sigma} \mathbf{q} \cdot \mathbf{n}$ of the convex surface $\Sigma = \{\mathbf{q} : \cosh q_x + \cosh q_y + \cosh q_z = \Lambda\}$ with $\Lambda = (3 \text{ den}/\text{num}) \cos \omega_b$; along an axis $\cosh \kappa = \Lambda - 2$, the law's line, along a face diagonal*

$\sqrt{2} \operatorname{acosh}((\Lambda - 1)/2)$ and along a body diagonal $\sqrt{3} \operatorname{acosh}(\Lambda/3)$ per unit of length; (iii) in integers the symmetry is exact.

Proof. (i) Without the rounding the line is $\mathbf{a}_{\text{next}} + \mathbf{a}_{\text{before}} = \mathbf{M}\mathbf{a}_{\text{now}}$; $\mathbf{M} + c\mathbf{I}$ for a large c is non-negative and irreducible (every $R_a/w > 0$ inside the guard, the GameBoard connected through its Links), so by Perron and Frobenius its largest eigenvalue is simple with a strictly positive eigenvector, the bound mode above the band. The well being invariant, $\mathbf{M}$ commutes with every g of the 48; $\mathbf{a} \circ g$ is then an eigenvector of the same eigenvalue, so $\mathbf{a} \circ g = \pm \mathbf{a}$ by uniqueness, and $+$ by positivity. (ii) Outside the well the mode solves the vacuum's line at the rotation ω_b above the band; every such solution is a sum of evanescent waves $e^{-\mathbf{q} \cdot \mathbf{x}}$ with $\mathbf{q}$ on Σ (Proposition 1 with $k = iq$ on each axis), and far away in the direction $\mathbf{n}$ the slowest one, $\min \mathbf{q} \cdot \mathbf{n}$, dominates; Σ is convex and symmetric, so the minimum lies at $q_y = q_z = 0$ for an axis and at the symmetric point for a diagonal (DERIVED: the full proof is the saddle point of the lattice Green's function). (iii) The generator is a deterministic function commuting with every g , so a symmetric seed stays symmetric bit for bit. $\square$

Checked on a lattice of 41^3 Nodes with the count 4,000 at one Node, $\Gamma = 12,000$, the pair $[2, 3]$ and the level entering the Link twice, from a random, unsymmetric seed (COMPUTED, the script beside the paper's source): $\omega_b = 0.6578$, the law's own table; positive everywhere; the largest asymmetry under the 48 2.8×10^{-17} of the peak; with the far field's $1/r$ removed, the decay along an axis 1.018 against 1.015, along a face diagonal 1.4650 per step against 1.4651, along a body diagonal 1.806 per step against 1.807; the level set at 10^{-4} of the peak lies 7.00 Links out along an axis and 6.93 along a body diagonal. So the bound record's tail is near a sphere, bulging by about one percent along the axes at this depth, exactly a sphere in the limit of the edge ($\Lambda \rightarrow 3$), and never the octahedron of the loading form $bt^{|dx|+|dy|+|dz|}$, which decays 1.7 times too fast along the diagonals: that form is the seed, and the shape is Rule3's own output, independent of the seed by (i).

7.3 A hollow alone: the saddle

Proposition 9 (the scaling of a hollow). *Let a body of M quanta stand at the radius R under one held hollow of reach a and divisor E , with the band's pressure the mode's curvature. Its energy scales as $E(R) \sim M/R^2 - M^2 G_a(R)/E$ with $G_a(R) \sim 1/R$ for $R < a$ and $\sim a^2/R^3$ for $R > a$. For $R > a$ the pressure wins as R grows and the body disperses; for $R < a$ the well wins as R shrinks once M exceeds the scaling's critical mass, and the body collapses; the fixed point at the critical mass is a saddle, and no mass has a stable extended body under one hollow.*

DERIVED, the scaling of Choquard's functional on the lattice; the law's own line says it in words: a hollow alone in three dimensions binds no record, it collapses into its own well or disperses [1, the paces]. Two hollows of different reach are one hollow for this purpose.

The run's finding. A GameBoard reading of the engine on the universe at 6,000, reported as a finding and not as a result: the fixed point of the binding row on a box of $48 \times 24 \times 24$ Nodes is the saddle of Proposition 9. Bodies of 29,000 and 31,000 quanta diffuse to clouds, 32,000 and above collapse, and 29,500 to 30,000 stand at the band's edge, the rotation 1.345 against the top 1.333. Two such bodies close by 0.63 Links at 100 intervals and 5.2 at 200 against the blind expectation 0.14 and 0.55, and a control of one body alone on the same board drifts the same way; the difference, two bodies less the control, is 0.06 and 0.11 against the blind 0.10 and 0.17, below the noise. Newton's fall is neither confirmed nor refuted by that run: bodies at the band's edge have tails reaching the faces, and a spreading body drifts toward the board's centre faster than it falls.

What ends a collapse. In the law as it stands the guard ends it: the pace's floor $0 < p$ (Proposition 3). A collapsing body stops where its central Nodes reach the horizon and freeze, a cluster of horizon Nodes; the generator's refusal of a collapse is that floor, and the rule's heavy body is such a cluster and no ordinary matter.

Hypothesis 4 (the hill holds the body). A body stands as a stable extended fixed point where a hill, a held row read with the opposite sign (by q , like signs), has a reach different from the hollow's. Below the hill's reach both are Coulomb-like, and the hill, with the smaller divisor, wins and forbids the collapse; between the hill's reach and the hollow's the hollow wins and binds; so the fixed point is a minimum for every mass above the smallest that binds, and the body's size lies between the two reaches. The rows the law holds for it: the polarisation $[1, 2]$ read by q , the reach $1/\kappa = 0.49$ Links ($e^{-\kappa} = 0.127$), a hard core of one Link; the binding row, the hollow of reach two Links; gravity $[1, 1]$, the pull. The reverse order, a long hill of the charge's $[1, 1]$ and a short hollow, gives a body with a largest mass, as a nucleus.

The hypothesis is the law's own sentence, a body is a balance of a hollow and a hill of different ranges, made a criterion by the scaling of Proposition 9; a run of the generator with the third row decides it, and it is the open item on which every experiment of a bound body waits.

7.4 The universe is bound; the click joins and parts

Assumption 1 (the universe is bound). Every body is bound, no body is free, and a record that is in no bound body is a message between two clicks, the one that gave it and the one that will take it: a matter record that no well holds disperses; light never binds, its band's bottom is 0.

A fall is the bias of a body's clicks, its tail longer toward the well (the local band lower there, κ smaller), so the remainder crosses T first on that side. The assumption removes no number; where it fails (the universe of record, nature) the distances of Section 9 stay. *The click joins and parts:* two bodies whose tails overlap read each other by the count's line and click, the tail biased toward the deeper, so the smaller gives quantum by quantum to the larger until its count falls under what binds and it dissolves into messages the larger takes, one record for the joined body; a body parts where two tails cease to overlap, beyond the click's reach $\ln(a_1 a_2 / T) / \kappa$; the one record opening for the joined body and closing everywhere for the parted one is the half quantum's one non-local act, and there is no other; the conversion changes the family. *The observer is a body:* a Node reads six Ports and a body's record is what its Nodes carry, so inside a body everything is seen (a GameBoard reading is the local language of its own record) and beyond its tail only a whole quantum arrives, a click.

7.5 The cluster universe of the divisor 1, a hypothesis

Hypothesis 5 (the rule's own universe of clusters). A universe whose every number is the rule's: the pairs from the integers 1, 2 and 3 of the three exact rotations and the three axes, $[1, 1]$ the vacuum's band on which the two real fields stand, $[1, 2]$ the exact band, $[2, 3]$ the matter pair (the first pair past the exact bands, which the universe of record approximates by 6,667 over 10^4) and $[-1, 2]$ the third; every divisor 1, the well the count itself; Γ a multiple of 6 so that every pair is exact over it; T the one unit, cancelling in every reading: zero free numbers.

Under it a body that binds itself at rest is one Node: its count in $[0.2255 \Gamma, \Gamma \text{ div } 2]$, the lower edge where the well of one Node first holds a rotation below the vacuum's band (the Green's function of the six Ports, a pure number of the rule, COMPUTED) and the upper the horizon where the

Link's pace freezes ($p_a = \Gamma - 2c - c_a$, DERIVED); its tail evanescent by $\kappa = \text{acosh}(\frac{9}{2} \cos \omega_b - 2)$ per Link, its record loading with $b = \text{isqrt}(cT \text{ den div } (2 \text{ den} - a))$ for the clock pair $[a, \text{den}]$ of $2 \cos \omega_b$; a heavier body a cluster of such Nodes; a screen a sieve of them at the period 2, Bragg's mirror for the wavelength 4 that a body at the edge gives ($\cos k = 3 \cos \omega_b - 2 = 0$); the hierarchy of clusters recursive with the click as the one pointer between levels; the bound states one rung per integer amplitude, about 67 at $\Gamma = 6,000$ with the ratio of neighbours $1 + 2/b$, a near continuum; and the electron the lightest band (Section 6.5). The finding of the generator on it, a GameBoard reading: the count-well collapses every wide body to one Node, a cluster of 27 Nodes of 32,590 quanta to 6,288 in one Node in three rounds, and the count laid from the form at every period does not conserve the count (23 quanta to 8,175 in 300 intervals at $\Gamma = 24$). So that universe holds no extended body, and Proposition 9 says why: the count-well is one hollow. It stays as the hypothesis under its own name, replaced in the files by the universe of whole bodies; its consequences stand under it and under nothing else.

7.6 The Node clock is not constant

Hypothesis 6 (the Node clock grows). $\Gamma_{t+1} = \Gamma_t + 6$, the step the six Ports of a Node and no free number.

The rule's physics depends on the ratios c/Γ and m/Γ alone, so every pair steps with Γ ($[m_0 \Gamma_t \text{ div } \Gamma_0, \Gamma_t]$) and the wall, A and the twist's unit are derived from Γ_t each interval, the declaration holding the Node clock as $[\Gamma_0, 6]$ and the rule no number. A count is the integer its Node holds and stays, so a body that does not gather falls under the rising threshold of binding, the small giving to the large being the way a body keeps its c/Γ ; light on $[\Gamma, \Gamma]$ does not drift, the clocks of bodies do, a body of a fixed count slowing as c/Γ falls with the rate

$$H = \frac{6}{\Gamma} \left| \frac{d \ln \omega_b}{d \ln(c/\Gamma)} \right|, \quad (9)$$

the rule's Hubble rate, and $\Gamma = 6t$ reads the universe's age in intervals; a look reads it as the takers' clock ratio growing by $(6/\Gamma)(s_{\text{taker}} - s_{\text{giver}})$ per interval. A GameBoard closed on every axis has no rest of the held field under a net count other than 0, so a bound universe closed on itself carries a mean of the held field that drifts by the net count each interval: the hypothesis that the universe expands, under its own name and no look of today.

7.7 The tree

The root is the law in one line. From it alone: the exact bands, the three reads and their ranks, the edge P , the click's theorems, the shape of a symmetric well's bound record, and the scaling of a hollow. With the owner's word, whole bodies and two held rows: the generator's fixed point and the declaration of Section 7.1. From a run of that generator, a finding: the fixed point of one hollow is a saddle. Under the hypothesis in its own name, the hill holds the body; under the assumption that every body is bound, the click that joins and parts and the observer as a body; under the hypothesis of the divisor 1, the cluster universe with its edge, horizon, sieve, wavelength and ladder, which holds no extended body; under its own name, the growing Node clock. Not closed toward nature by name: the three distances $\alpha = 1/137.036$, m_e/m_P and m_p/m_e , the charges $2/3$ and $1/3$, the generations, the band $[0, 1]$, and now what holds a body against its own collapse.

8 What the law says of nature

8.1 The postulates

1. The world is Nodes and Events. Space is the GameBoard; an Event is a wall crossed by a remainder at a Node: a level stepped, a count moved, a click; nothing else happens.
2. A Node holds bounded local information: its NodeState, fixed in size, read from itself and its six neighbours; no read beyond them, no list that grows with the world, nothing of the past kept. Every physical update uses its own record and the six causally available neighbours with fixed work and storage for a fixed set of families; a self-field estimated or subtracted from anything global is forbidden, whatever the size of its final answer.
3. Consistency is local and causal: an Event first changes its own Node and travels Link by Link, one Link per interval, each Node updating on receipt under the same rule. The one exception is the click: a record ends at once, whole, at the detector.
4. The causal speed is one Link per interval, built in; every other speed is a rational, a count of Links over a count of intervals.
5. Every physical calculation is on bounded integers; there is no float, no root and no draw in the law; a Node keeps only the law's own numbers: each record's two levels and its remainder, the family's pair and the Node clock from the content there.
6. A measurement is a detector's click, an action of the law on the state: the record ends at the detector and the detector's own record changes. Only a click is compared with nature or pinned as an expectation; displays and the host's readings read state and write nothing.
7. The Inside is the GameBoard, where no one measures; the Outside is the detectors and their clicks, the only thing claimed to represent nature.

8.2 The constants

The speed of light is one Link per interval. The quantum of action is one click, its energy the quantum's action T . Newton's constant is not declared: a body of energy count s makes the level $c(r) = s \times \text{the reference flux}/E_s/r$ at the distance r in Links (the hold's row: the count enters the field's line over the row's divisor E_s), and the level slows the clock by the potential $\Phi = -c/\Gamma$, so

$$G = \frac{\text{the reference flux}}{\Gamma E_s} \text{ per unit of energy count,} \quad (10)$$

with the reference flux a number of the GameBoard's geometry (about 1.4 for a cube of side 3) and the divisor E_s the energy unit setting the mass of one unit of level [1, the constants]. A world at a smaller Γ has stronger gravity per unit of content; the ratios between rows carry G once and cancel it. The quantum of distance is the Link, of time the interval; nothing between two Nodes or two intervals is observed. The potential at a body's Nodes changes by G per unit of energy count and never by less: the quantum of gravity is the click.

The three free numbers and their bounds. The cluster universe of the divisor 1 (Section 7.5) carries none of the three; the universe of whole bodies declares two divisors and a binding pair; the universe of record carries all three, and its three are these. E_s of gravity is fixed by no line of the law, as nature's G is fixed by none (Rule3 is linear and its form's scale is free); its bound is the law's: at every Node of every body the pace stays positive, $\text{count} \times (1 + K/E_s) < \Gamma$ with K the hold's kernel summed over the body (about 51 for a body wider than the reach), so $E_s > K \times \text{count}/(\Gamma - \text{count})$ at the world's heaviest Node, and the reading's resolution bounds it above. The charge's E_s is free as α is, bounded below by the same pace at the bound charge's Nodes and by the train (one quantum

at least one period). The matter pair is free as the mass is, bounded by the band: $\text{den} > \text{num}$, and the reach $\cosh \kappa = 3 \text{den}/\text{num} - 2$ fitting the body's width.

8.3 The forms of nature the line reaches

Each form below is derived from the line before any run and is a prediction of the law, never a result: the reading named with it is what a detector's click would read, and no such reading is made in this paper. The form of nature beside it is on the comparison side and is never an input [1, the rows against nature]. $U = c/\Gamma$ is the potential at the point named, the clock's shift per held level (Proposition 2); u is light's group velocity at the wave number in question.

- (a) *The redshift* (DERIVED). By Proposition 2 at $k = 0$ a clock at the held level c rotates at ω with $\sin(\omega/2) = (1 - U) \sin(\omega_0/2)$, so two clocks of one family at the levels c_1 and c_2 have their rotations in the ratio $(1 - U_1)/(1 - U_2)$, $1 - (U_1 - U_2)$ to first order, the deeper clock the slower; the law's reading through the Link's pace is the root of $(1 - 2U_1 + 2U_1^2)/(1 - 2U_2 + 2U_2^2)$. Beside it: Einstein's gravitational redshift, $1 - \Delta\Phi/c^2$.
- (b) *The bending* (DERIVED). Along a Link light's speed falls by the level twice, once in the clock and once in the Link (Section 2.5), so past a body at the closest distance b the wavefront turns toward the body by twice the clock's share, and at a receiver L Links beyond the centroid is shifted by $4U_b L$. Beside it: Einstein's $4GM/(c^2 b)$, twice Newton's.
- (c) *The delay* (DERIVED). A round trip past the body is delayed by $4U_b$ times the path's length within the well's reach, the same two factors on the two ways. Beside it: Shapiro's delay.
- (d) *The perihelion* (DERIVED). An orbit at the semi-axis a and eccentricity e advances $6\pi U_p$ per turn, U_p the potential at $a(1 - e^2)$, the same two factors in the orbit's equation. Beside it: Einstein's $6\pi GM/(c^2 a(1 - e^2))$.
- (e) *The moving clock* (DERIVED). A body moving at v carries the phase k per Link on its bound band $\omega(k) = \omega_0 + \Delta(1 - \cos k)$, Δ its packet's top velocity, and its clock at the moving centre is $\Omega(k) = \omega(k) - k\omega'(k)$, the phase rate less k times the group velocity $v = \omega'(k)$; to second order $\Omega = \omega_0 - \Delta k^2/2$ and $v = \Delta k$, so $\Omega/\omega_0 = \sqrt{1 - v^2/c_b^2}$ with $c_b^2 = \Delta\omega_0$, read as the mean of the front and back click periods at two resting detectors. Beside it: Lorentz's factor, at the body's own light speed c_b and not the lattice's, since no boost is among the 48.
- (f) *The charge* (DERIVED). Like signs a hill, unlike a hollow; a reader of charge q reads $+qd$ in its pace and $-q(\mathbf{n} \cdot \mathbf{A}) \text{div } W$ on its vector part (the induction, Section 4), so like charges moving together repel less. Beside it: Coulomb's force and the magnetic force.
- (g) *Born, Malus, Bell* (THEOREM). The shares of a click are the current's shares (Theorem 6); a polariser's turn of the record's pair gives $\cos^2$ and $\sin^2$ of the angle between the pair and its axis; the pair record gives $\cos 2(a - b)$ and $2\sqrt{2}$ (Proposition 4). Beside them: Born's rule, Malus's law, Aspect's correlations and Tsirelson's bound.
- (h) *De Broglie's relation* (DERIVED). A giver of rotation ω_b radiates a family whose band holds it, $\text{num}/(3 \text{den}) \leq \cos \omega_b \leq \text{num}/\text{den}$, with the wave number from $\cos \omega_b = \frac{\text{num}}{3 \text{den}} (2 + \cos k)$ and the group velocity $v = \frac{\text{num}}{3 \text{den}} \sin k / \sin \omega_b$; to second order $k = \omega_b v / c_m^2$ with $c_m^2 = \omega_0 / (3 \tan \omega_0)$, the family's own light speed. Beside it: $\lambda = h/p$ with the rest rotation as the mass.
- (i) *Interference* (DERIVED). A record split at two openings d apart meets on a screen L beyond with the two arms' phases $\mathbf{k} \cdot \mathbf{r}$: the shares $\cos^2(\pi dy/(\lambda L))$ under one opening's envelope to first order, the bright strips $\lambda L/d$ apart, and the lattice's dispersion beyond the first order (COMPUTED: at $\lambda = 4$ on $d = 16$ and $L = 97$ the first bright pair moves from 6.1 to 9.3 strips of 4 Nodes, Table 1); two arms rejoined at a splitter of share s give $4s(1 - s) \cos^2(\Delta/2)$ at the cross exit, Δ the arms' phase difference. Beside them: Young's and Mach-Zehnder's fringes.

- (j) *The fall* (DERIVED). In a pace gradient a bound packet accelerates at $a = \Delta |d\omega_0/dc| (c_+ - c_-)/D$ toward the well (Section 4), so bodies with the same $\Delta \times d\omega_0/dc$ fall alike, and across the bound bodies of one family that product differs by the band's own shape, a tenth (Section 9). Beside it: Galileo's fall and the equivalence principle.
- (k) *The pull and the orbit* (DERIVED). The held field on $[1, 1]$ rests at Poisson's with the sources and at Laplace's outside a body, $c(r)$ falling as $1/r$, so the acceleration falls as $1/r^2$ and a circular orbit's v^2 as $1/r$. Beside it: Newton's pull and Kepler's fall-off; a flat $v(r)$ would name a missing law.
- (l) *The medium* (DERIVED). Inside a body of pace p light of wave number k is slowed to $v_{\text{in}} = (p/\Gamma)^2 \sin k_{\text{in}} / (3 \sin \omega)$ with $1 - \cos k_{\text{in}} = (\Gamma/p)^2 (1 - \cos k)$ (Eq. (5)), the index u/v_{in} , and is reflected whole where $p < \Gamma \sin(k/2)$; through a body moving at w the wave number is matched at the moving face, $\omega - kw = \omega_{\text{in}} - k_{\text{in}}w$, and the speed inside is the band's group velocity there. Beside them: the refractive index and the mirror; Fresnel's drag $w(1 - v_{\text{in}}^2/u^2)$ stands beside the law's number and is not derived from it.
- (m) *Doppler and Sagnac* (DERIVED). A giver receding at v from a mirror at rest reads its returned clicks every $P(u+v)/(u-v)$ intervals for a giving every P ; a giver moving at v along a ring of N Links reads its two arms $2Nv/(u^2 - v^2)$ apart, $4A\omega/u^2$ for the ring's area A and angular speed ω to first order. Beside them: the two-way Doppler ratio and Sagnac's $4A\omega/c^2$, with u in place of c .
- (n) *A free record keeps its wave number* (THEOREM). Rule3 is translation-invariant, so a free record keeps its wave number and rotation exactly and a giver's period is read unchanged at every distance, the rounding's walk at most one unit of level per Node per interval. Beside it: Hubble's drift, for which the law's own line is the growing Node clock of Hypothesis 6.

8.4 The experiments run algebraically

Each experiment of the law's rows is run here algebraically: a world is declared in numbers, and the reading a detector would give is computed from the formulas of Section 8.3 and of the law, by no run of the engine (COMPUTED; the script beside the paper's source computes every number of Table 1). The universe is the universe of record: $\Gamma = 10^4$, light $[1, 1]$, the matter pair 6,667 over 10^4 ($\omega_0 = 0.8410$, $\Delta = 0.2982$, $c_m^2 = 0.2508$), a second matter pair 8,000 over 10^4 where two families are compared; $U = c/\Gamma$; light of $\lambda = 4$ has $u = 0.4472$. The bands are the rounding's, one Node or one interval. Where the law's number differs from the form of nature at the declared numbers, the difference is the law's own, read blind.

Table 1 The experiments run algebraically on declared worlds (COMPUTED): the reading the law gives, and the form of nature at the same numbers.

Row	The declared world	The reading computed from the law	The form beside it
(a) the redshift	two clocks of one family at the levels 500 and 1,000 ($U = 0.05$ and 0.10)	the tick ratio: the one-Node clock of Proposition 2, $(1 - U_1)/(1 - U_2) = 1.0556$; the law's row through the Link's pace 1.0506; the first order 1.0500	Einstein's first order 1.0500; Schwarzschild's 1.0607; the second order is the law's own
(b) the bending	$U_b = 0.02$ at $b = 20$ Links, a receiver 200 Links beyond	the centroid 16 Nodes toward the body, the band one Node	Einstein's 16; Newton's 8
(c) the delay	$U_b = 0.02$, 100 Links within the reach	8 intervals on the round trip	Shapiro's 8

Row	The declared world	The reading computed from the law	The form beside it
(d) the perihelion	$U_p = 0.01$	0.1885 rad, 10.80 degrees per orbit	Einstein's 10.80 degrees
(e) the moving clock	the matter pair at $v = 0.1$ ($c_b = 0.5008$)	$k = 0.3420$; the factor $\Omega/\omega_0 = 0.9799$	Lorentz's at c_b : 0.9799; at the lattice's $1/\sqrt{3}$: 0.9849
(f) the charge	a unit source, readers at $r = 10$ and 20 Links	the levels 200 and 100, Laplace's $1/r$; the force ratio 4; the moving term's coefficient is not computed here	Coulomb's $1/r^2$
(g) the two slits	$\lambda = 4$, $d = 16$, $L = 97$	the first bright pair at 37.2 Nodes (9.3 strips of 4), the second at 71.8, by the stationary phase on the lattice's dispersion	Young's $\lambda L/d = 24.25$ Nodes (6.1 strips): the lattice moves the strips
(h) Bell	the settings 0, $\pi/4$ and $\pi/8$, $3\pi/8$	$E = 0.7071$, -0.7071 , 0.7071 , 0.7071 ; $S = 2.828$; the marginals 0.5	Aspect's; Tsirelson's 2.828; the local bound 2
(i) the fall	a tent of slope 2 levels per Link, a fall of 50 Links, the two matter pairs	$a = 5.33 \times 10^{-5}$ and 5.93×10^{-5} Links per interval squared; the fall in 1,369 and 1,299 intervals	Galileo's: alike; the law's two families differ by a tenth, the band's own shape
(j) de Broglie	a giver of $\cos \omega_b = 0.60$ on the matter band	$k = 0.7956$ ($\lambda = 7.90$ Links), $v = 0.1984$	de Broglie's second order $k = 0.7337$, 8 percent off at this v
(k) the two-qubit computer	a control of amplitude 100 at $g_c = 1$	the target's pace shifted by 0.25 of a level; the phase 0.022 rad over 1,000 intervals	a controlled phase of order one: none
(l) Mach-Zehnder	a one-Node splitter at the counts 1,000, 2,000, 2,929 and 4,000 for $\lambda = 4$; equal arms	the reflected share 0.052, 0.240, 0.500 and 0.760, one half exactly at the mirror's count $\Gamma(1 - \sin(k/2))$; the cross exit 1.000 at equal arms, 0.500 at a quarter turn, 0 at a half	the interferometer's fringes
(m) the round trip	a giver receding at $v = 0.05$, giving every 8 intervals at $\lambda \approx 4$	$k = 1.417$, $u = 0.473$; the returned clicks every 9.89 intervals	the two-way Doppler ratio 1.236 with u for c
(n) Sagnac	a ring of 400 Links, $v = 0.05$, $\lambda = 4$	the arms return after 804.5 and 1,007.0 intervals, 202.5 apart	Sagnac's $2Nv/(u^2 - v^2) = 202.5$ with u for c
(o) the moving mass	the record of (j) toward a set 100 Links away, the giver of M quanta	the clicks 504 intervals after each giving; the giver's velocity moves by $0.221/M$ per quantum	momentum conservation
(p) the medium	a window of level 1,000 and depth 20 for $\lambda = 4$; the window moving at $w = 0.05$	$k_{\text{in}} = 1.808$, $v_{\text{in}} = 0.352$, the index 1.270, the passage 12.1 intervals longer; in the moving window $k_{\text{in}} = 1.847$ and the speed inside 0.344	the index; Fresnel's 0.371 against the law's 0.344: a departure
(q) dark matter	a heavy body's level 200 at $r = 10$, a test body on circular orbits	the level 100 at $r = 20$; $a = 5.3 \times 10^{-4}$ and 1.3×10^{-4} ; $v = 0.073$ and 0.052, the ratio $1/\sqrt{2}$; the periods 860 and 2,433 intervals	Kepler's fall-off; a flat $v(r)$ would name a missing law

Row	The declared world	The reading computed from the law	The form beside it
(r) dark energy	a giver of period 8, sets at 1,000 and 10,000 Links	the mean click interval 8.00 at both; the packet of width 10 spread to 20 and 200 Links, its count unmoved	Hubble's drift: none without Hypothesis 6

8.5 How the forms are reached, not assumed

The law holds none of the forms it is compared with. The causal speed is one Link per interval by construction, and light's group velocity $1/\sqrt{3}$ at long wavelength is Proposition 1's, not a constant declared. Planck's constant is the click's unit T , and $E = h\nu$ is the record's click rate against its rotation. Newton's constant is Eq. (10), a divisor and the Node clock; the factors 4 in the bending and the delay and 6π in the perihelion arise from the level entering the Link twice and the clock once; Lorentz's factor is the dispersion of the body's own bound band read at its moving centre, with the body's own light speed and not the lattice's; de Broglie's relation is the band's group velocity; Born's rule, Malus's law and Tsirelson's $2\sqrt{2}$ are the ladder's and the pair record's theorems. Each form stands beside the law's own expectation, and where the two differ the difference is written before any run, as in the rows (a), (g), (i), (j) and (p) of Table 1.

9 What is open, and what is not claimed

The law's document keeps its open points by name [1, what is open]:

1. Whether a body at rest jitters when its current's swing reaches $T/2$.
2. The self-source's cubic term: not in the law, the line is the squares' sum alone.
3. The twist table's small angles: a triple with d at most 10^9 reaches no angle below 6.3×10^{-5} radians.
4. The equivalence principle holds to the band's own shape: $\Delta \times 2 \tan(\omega_0/2)$ differs by a tenth across the bound bodies of one table; the fall with two light bodies reads the rest.
5. The two thirds' charges: a third with the sign q reads a third, and the composite read modulo a turn identifies $-1/3$ with $+2/3$, but no line gives the up third $2/3$ and the down third $1/3$.
6. The three generations: a hypothesis under its own name, the three ranks of the exact band $[-1, 2]$; no line gives a rank a mass.
7. The exact band $[0, 1]$ of the period 4: a family that never clicks and carries no colour, unnamed in nature here.

And three of this paper's own, findings and no lines: (8) what holds a body against its own collapse, Hypothesis 4 until a run of the generator with the third row decides it; (9) the fixed point of one hollow is a saddle, and the count laid from the form at every period does not conserve the count (Section 7.5); (10) the count's line's inverse: the whole-or-nothing block of a starved Node leaves no trace that the backward step can read, so the whole-run reversibility through the count's line does not stand, while the step between clicks is exact (Theorem 1). Beside them stand the three distances of Section 7.7, α , m_e/m_P and m_P/m_e , of which the last fixes the atomic level's Γ and is not predicted, and the spin $1/2$.

Not claimed: any agreement with nature, since nothing is run here; the Lorentz group, which is no symmetry of the lattice; a controlled phase of order one between two records, which Theorem 2 excludes from the acts alone; the count of gravitational levels, which is the universe's history; the reach of the divisor 1 in the universe of record; and Newton's fall, which the one run reported neither confirms nor refutes. Every hypothesis of the paper is named as one, and the tree of Section 7.7 says what each claim needs beyond the line.

Declarations

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Data and code availability. The law’s document, the simulator that computes the rule and the figures’ scripts are archived at Zenodo under the concept DOI 10.5281/zenodo.22738746 [17], under the MIT licence; no run of it stands in this paper.

Author contributions. The author laid down every postulate and hypothesis of the model, directed the work and wrote the paper.

The use of a large language model. The author used a large language model (Claude, Anthropic, through Claude Code, 2026) extensively as a tool, under his direction and instructions, for the software, the computations, the mathematical derivations and their checks, the physics of the law’s document and the drafting of the text; the model is not an author; the author reviewed and edited every statement and takes full responsibility for the content. No figure was made by a generative tool; every figure is drawn by a script from the definitions.

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A Where this model stands among discrete models

Deterministic lattice models of physics are old: the lattice gases of Hardy, de Pazzis and Pomeau [8] and of Frisch, Hasslacher and Pomeau [9] recover hydrodynamics from Boolean collisions; the quantum cellular automata of Bialynicki-Birula [10], Meyer [11] and Arrighi [12] step a unitary state on a lattice. This model differs from both in its state and in its measurement: the state is integer levels with a remainder per Node and no complex amplitude, and the measurement is a division of the law and no projection. It differs from the pilot-wave account [13, 14] in that no particle position is kept beside the wave: the count follows the record’s current by the count’s line, and a body’s position is its count’s centroid. It differs from spontaneous-collapse models [7] in that the click’s residue is the rule’s own remainder and no stochastic term. In the ontological-models language [15, 16] the record is the ontic state and the click its one read; the correlation $2\sqrt{2}$ is reached, and Popescu and Rohrlich’s 4 [6] is not, since the second label’s shares are a rotation’s squares.